ARBITER Direct support | Context | Inference

Unassisted
same prompt, no ARBITER

D1 **Low**

1.2 … concealed until participants were enrolled …? **Yes**

Direct support · CIRCULATIONAHA.121.057639.pdf · p 3 · L46–50 · 592b9a6c53

… web-based interactive response system. The assignment to each treatment group was computer-generated and … by means of a permuted-block sequence with variable block size.

(1) **Bound span**

Justification: Patients were randomized using a web-based interactive response system…

Unknown: Whether the central system was run by a unit independent of enrolling staff is not stated explicitly.

(4) **Recorded gap**

1.1 Y · 1.2 Y · 1.3 PN

Allocation used a computer-generated, centre-stratified permuted-block sequence delivered through a web-based interactive response system, so concealment was adequate.

D2 **Low**

2.1 Were participants aware of their assigned intervention …? **Yes**

Direct support · ClinicalTrials.gov NCT02436655 · p 1 · L67–68 · d32e23eb2a

Masking: NONE

Justification: The registry lists masking as NONE…

2.4, 2.5, 2.7 not asked — RoB 2 branching

(2) **Branching**

2.1 Y · 2.2 Y · 2.3 PN · 2.4 NA · 2.5 NA · 2.6 Y · 2.7 NA

The trial was open label, so participants and clinicians knew the assignments.

D3 **Low**

3.1 Were data … available for all, or nearly all, participants …? **Yes**

Direct support · ehae585.pdf · p 6 · L26–32 · a890f771fd

All patients were included in the analysis. … One patient, who was randomized to early surgery, has been lost to follow-up, and this patient was censored…

Justification: so 1/157 (0.6%) has incomplete outcome data.

3.1 Y · 3.2 NA · 3.3 NA · 3.4 NA

All 157 patients were analysed. One patient lost to follow-up was censored.

D4 **Low**

4.3 Were outcome assessors aware of the intervention received …? **Yes**

Direct support · ehae585.pdf · p 3 · L1–116 · a890f771fd

Data and safety monitoring board members were not blinded to the treatment allocation during events review.

Justification: the report states that DSMB members were not blinded to allocation when reviewing events.

4.4 … influenced by knowledge …? **Probably no**

All-cause death is an observer-reported outcome that involves no judgement…

4.1 N · 4.2 N · 4.3 NA · 4.4 NA · 4.5 NA

Death is an objective outcome. Assessors were not blinded to allocation, but this cannot plausibly bias an all-cause death count.

4.3 skipped despite this → invalid answer path

D5 **Some concerns**

5.1 … pre-specified analysis plan … before unblinded outcome data …? **No information**

Context · CIRCULATIONAHA.121.057639.pdf · p 4 · L16–35 · 592b9a6c53

Prespecified secondary end points included the following: … Time to death

Context · ClinicalTrials.gov NCT02436655 · p 1 · L95–101 · d32e23eb2a

Primary outcome 1: all cause death, Major Adverse Cardiac Event (MACE) including: …

Justification: names all-cause death only as a component of composite outcomes.

No dated source therefore specifies this measurement and analysis.

5.1 NI · 5.2 PN · 5.3 PN → official algorithm

(3) **Algorithm**

Proposed Some concerns → Adopted Some concerns (no override)

5.1 PY · 5.2 PN · 5.3 PN

Time to death was a prespecified secondary endpoint with a registered 5-year time frame.

Registry lists death only within a composite (cf. ARBITER 5.1)

Overall **D1–D4 Low · D5 Some concerns →** **Some concerns**

Official RoB 2 overall rule: any Some concerns; applied by the server

Bundle verifier: verified

Published: Low **Adjudicators: Some concerns** **Unassisted: Low**